A RESOLVING SET OF OBJECTS COMPUTES THE DERIVED FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5336-5344, following the statement at 5308-5334.

Editorial note. The final step reads the definition of RF defined at X (essentially constant ind-object) directly; the dual half is the mirror argument.

Allowed standard prerequisites (editorial).

- Definition of defined at and computes. definition of RF at X via the essentially constant ind-object (same chapter, at ~4764). Essentially constant with value F(X') means RF defined and F(X') the value.

Source reference. Original reference 05S8: sources/stacks-derived/pinned-original-source.tex, source lines 4736–4736. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Assumptions and notation as in Situation 05S8. If there exists a subset $\mathcal{I} \subset \mathrm{Ob}(\mathcal{D})$ such that*

(1) *for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X \to X'$ in S with $X' \in \mathcal{I}$, and*
(2) *for every arrow $s : X \to X'$ in S with $X, X' \in \mathcal{I}$ the map $F(s) : F(X) \to F(X')$ is an isomorphism,*

then RF is everywhere defined and every $X \in \mathcal{I}$ computes RF. Dually, if there exists a subset $\mathcal{P} \subset \mathrm{Ob}(\mathcal{D})$ such that

(1) *for all $X \in \mathrm{Ob}(\mathcal{D})$ there exists $s : X' \to X$ in S with $X' \in \mathcal{P}$, and*
(2) *for every arrow $s : X \to X'$ in S with $X, X' \in \mathcal{P}$ the map $F(s) : F(X) \to F(X')$ is an isomorphism,*

then LF is everywhere defined and every $X \in \mathcal{P}$ computes LF.

Proof. Let X be an object of $\mathcal{D}$. Assumption (1) implies that the arrows $s : X \to X'$ in S with $X' \in \mathcal{I}$ are cofinal in the category X/S. Assumption (2) implies that F is constant on this cofinal subcategory. Clearly this implies that $F : (X/S) \to \mathcal{D}'$ is essentially constant with value $F(X')$ for any $s : X \to X'$ in S with $X' \in \mathcal{I}$. □

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